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"24CS310"

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester: **Test II** **Section : C,D and E**

Object Oriented Programming using C++

Duration: 1.15 Hr Date: 07/01/2026 Max. Marks: 30 Marks

NOTE: Part A Question compulsory.

PART A

Q.No.	CO	CD	PI	QUESTION	MARKS
1 (a)	CO2	L3	2.1.2	Design a C++ class to represent a complex number using real and imaginary parts. Implement constructor overloading and operator overloading to add two complex numbers using the '+' operator. Demonstrate the working of the program by displaying the operands and the resultant complex number.	05
1(b)	CO2	L3	2.1.2	Design a class Counter with an integer member count. Implement the **prefix ++ operator** so that when it is applied to a Counter object, the count is incremented by 1. Write a program to show the operator in action.	05

1(c)	CO4	L3	2.1.2	Write a program which reads a text from the keyboard and displays the following information on the screen in **two columns**:
				(a) Number of lines
				(b) Number of words
				(c) Number of characters
				Note: Strings should be **left-justified** and numbers should be **right-justified** in a suitable field width.
				Output as follows:
				Output
				`Enter text (type 'END' on a new line to finish):`
				`This is a test`
				`Welcome to C++ programming`
				`Have fun coding`
				`END`
				
				`Description Count`
				`----------`
				`Number of lines 3`
				`Number of words 11`
				`Number of characters 55`
					05

2a	CO3	L3	2.1.2	Write a C++ program to define a class Student with data members name and rollNumber. Include a member function display() to print the student details. In the main() function, create an object of the Student class and use a **pointer to the object** to	07

				assign values to the data members and to call the display() function.	
2b	CO3	L3	2.1.2	Write a C++ program to demonstrate the use of the **this pointer**. Define a class Counter with an integer data member count. Implement a member function increment() that increments count and returns the current object using the this pointer. Also, include a function display() to show the value of count. In the main() function, create an object of the Counter class, call the increment() function multiple times, and display the final value.	08
OR					
3a	CO3	L2	1.4.1	Explain multilevel inheritance in C++. Mention any two advantages of multilevel inheritance. Write a C++ program to demonstrate multilevel inheritance using three classes, where a derived class is created from another derived class.	07
3b	CO3	L2	1.4.1	Explain the following terms in C++: 1. Pointer to object 2. this pointer (Write brief explanations with examples where necessary.)	08

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Understand object-oriented programming concepts and be able to design solutions for a given problem.	L3
CO2	Able to reuse code with extensible Class types, user defined operators and function overloading.	L3
CO3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism	L3
CO4	Incorporate file-handling functions and exceptions for a given application.	L3
CO5	Design C++ packages with templates for providing	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying

--- End ---